

Aswin VATTAPPARAMBATHU JAYAPRAKASH

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EXPERIENCE

XR Research and Development Intern

LIRIS – CNRS Laboratory • Saint-Étienne, France

Jun 2025 - Sep 2025

- Link: <https://github.com/liris-xr/6dof-vr-thermal-feedback>
- Extended an open-source VR thermal feedback framework (Unity C#) by implementing obstacle-aware thermal occlusion, increasing environmental realism for immersive simulations.
- Integrated haptic vest with custom vibrotactile patterns, expanding the system from thermal-only to multisensory interaction.
- Designed and deployed a multisensory VR road-crossing simulation (thermal + haptics + spatial audio); user testing reported higher immersion compared to baseline VR.
- Improved internal developer tooling (visual debugging, teleport navigation, thermal source tagging), accelerating prototype iteration for the research team.

EDUCATION

M.Sc. in Imaging and Light in Extended Reality (IMLEX, Erasmus Mundus)

Univ. of Eastern Finland | Univ. Jean-Monnet | Toyohashi Univ. of Technology • Finland | France | Japan

Sep 2024 - Sep 2026

- Joint master's degree combining XR, computer vision, photonics, and robotics across three institutions
- Thesis: "Adaptive XR interface for Digital Twin manipulation" at National Institute of Advanced Industrial Science and Technology (AIST), Japan

B.Sc. (Honors) in Physics

Hindu College, University of Delhi • Delhi, India

Nov 2020 - Jul 2023

PROJECTS

Feature Detection, Matching & Camera Calibration | OpenCV, Python

Dec 2025 - Dec 2025

Toyohashi University of Technology, Japan

- Developed full camera calibration and distortion correction pipeline using OpenCV, enabling accurate intrinsic parameter estimation and reprojection validation.
- Implemented robust feature-matching workflow to improve correspondence stability in low-texture and noisy environments.
- Analyzed feature failure cases and matching robustness to inform perception reliability in XR/vision-based systems.

VR Perception Study with Physical Props | Unity, XR Interaction Toolkit, OpenXR

Nov 2025 - Nov 2025

Toyohashi University of Technology, Japan

Link: https://aswin-avj.github.io/portfolio/projects_page/PlankExp/PlankExp.html

- Built a modular Unity-based XR research framework to evaluate physical prop influence on height perception.
- Engineered randomized experimental trials (40 trials/participant) with condition balancing and automated data logging.
- Observed measurable reduction in perceived height threshold (PSE) and increased cautious movement time with physical props.

360° Image Processing & Real-Time VR Visualization | CUDA, OpenCV, Three.js

Apr 2025 - Apr 2025

Jean Monnet University, France

Link: https://aswin-avj.github.io/portfolio/projects_page/360ip.html

- Developed GPU-accelerated equirectangular ↔ cube map conversion pipeline using custom CUDA kernels and OpenMP.
- Optimized panoramic projection and face-aware ray sampling for real-time performance.
- Built immersive WebXR viewer with depth-based displacement for enhanced 360° VR visualization.

*Additional projects and professional experience available at:

aswin-avj.github.io/portfolio

SKILLS

Programming: Python, C++, Unity (C#), Scilab (Matlab alternative)

XR & Graphics: OpenXR, XR Interaction Toolkit, Meta Quest SDK, WebGL, Unity, WebXR, CUDA, Three.js

AI & Vision: PyTorch, OpenCV, ROS2, Gazebo, Keras

Tools: Git/GitHub, LaTeX, DaVinci Resolve